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The Editor-in-Chief
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Dear Editor,

I am submitting for consideration the manuscript “**Metabolic Buckling of the Growing Spine**” for publication in *Spine Deformity*.

What the manuscript addresses. Adolescent Idiopathic Scoliosis (AIS) affects approximately 3% of adolescents worldwide, yet after five decades of research no single unifying causal mechanism has been established. Current models explain correlates of AIS (genetics, growth velocity, proprioceptive deficits) but do not quantitatively predict *why* deformity initiates in a narrow developmental window and why it preferentially affects the thoracic spine at T8–T10.

The central hypothesis. This manuscript introduces the **Derivative Gain Trap** — a first-principles, biomechanically grounded mechanism derived from Delay Differential Equation (DDE) control theory. During the adolescent growth spurt, rapid spinal elongation increases absolute neural conduction delay τ faster than the central nervous system can compensate. When τ exceeds a critical threshold τ_c , the derivative feedback gain K_D — which stabilizes posture during childhood — crosses a supercritical Hopf bifurcation boundary and becomes actively destabilizing. This transition, compounded by a concurrent Energy Deficit Window in which gravitational demand ($\sim L^4$) outpaces intervertebral disc nutrient supply ($\sim L^2$), produces the localized, coronal-plane instability characteristic of AIS.

Key quantitative results directly relevant to the readership of *Spine Deformity*.

- The critical spinal length $L_{\text{crit}} \approx 0.35$ m, at which metabolic demand exceeds supply capacity by 41%, retrospectively maps onto the 11–12 year age window of peak AIS incidence without any parameter fitting to clinical data.
- Buckling localization emerges naturally at spinal position $s/L \approx 0.55$ – 0.65 (T8–T10 levels) from boundary conditions alone (fixed pelvis, semi-free rib cage), consistent with the most common clinical presentation of right thoracic AIS.
- Bracing is modelled as derivative-gain augmentation that restores the stability margin, providing a biophysical rationale for the BrAIST trial outcomes (Weinstein et al. 2013 NEJM).
- Cross-species Bio-Gravitational Number analysis ($n = 12$ species, $R^2 = 0.744$) confirms that humans alone occupy the “Allometric Trap”, explaining AIS as a uniquely human condition.

Clinical implications. The framework generates specific, testable predictions including: (1) longitudinal proprioceptive latency measurements as a pre-deformity biomarker; (2) NAD⁺ precursor supplementation (e.g., Nicotinamide Riboside) during Peak Height Velocity as a pharmacological curve-modification strategy; and (3) the identification of high- K_D neuromotor phenotypes as a risk-stratification tool.

Originality and fit. This is the first report to derive the AIS initiation window from first-principles stability analysis of a delayed control system, validated against independent clinical datasets (Cheng 2015, Sanders 2017, Weinstein 2008/2013, BrAIST). The manuscript is not under consideration elsewhere. All simulation code is openly available at <https://github.com/sayujks0071/life>.

I declare no conflicts of interest. This study involves no human subjects or animal experiments requiring ethical approval; all data are computational simulations and published clinical datasets.

I believe this manuscript offers a mechanistic breakthrough that will be of significant interest to the clinical and research readership of *Spine Deformity*. I welcome the opportunity for peer review.

Sincerely,

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